

CANONICAL CIRCUIT DECOMPOSITIONS OF PARITY VECTORS AND THE FINITENESS OF PARADOXICAL COLLATZ SEQUENCES

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ABSTRACT. For the Collatz function T , Rozier and Terracol call a finite sequence $\Omega_j(n) = (n, T(n), \dots, T^j(n))$ with $n > 2$ *paradoxical* when its coefficient $C_j(n) = 3^q/2^j$ is less than 1 and yet $T^j(n) \geq n$. We show that the parity vector of any such sequence decomposes canonically into *circuits* — blocks of the form $1^a 0^e$ — so that the paradoxical set is exhausted by the family $(\mathcal{P}_R)_{R \geq 1}$ of its subsets consisting of the sequences whose parity vector spells exactly R circuits. We then prove that the existence of a bound on the length j , valid on $\mathcal{P}_R$ *uniformly in R* , is equivalent to the finiteness of the paradoxical set; by Corollary 3.3 of Rozier–Terracol such a bound would therefore imply the Collatz conjecture. This makes precise the sense in which the uniform-over- R strengthening of the per- R finiteness theorem is not merely sufficient for, but a reformulation of, the finiteness of the paradoxical set. All results are elementary and machine-verified in Lean 4.

1. NOTATION AND BACKGROUND

We follow the notation of [1] throughout. Let T be the Collatz function

$$T(n) = \begin{cases} \frac{3n+1}{2}, & n \text{ odd}, \\ \frac{n}{2}, & n \text{ even}, \end{cases}$$

on the positive integers, and for $j \geq 0$ let $\Omega_j(n) = (T^k(n))_{k=0}^j$ denote the sequence of iterates of length $j+1$ [1, Def. 4.1]. Following [1, Def. 2.1], $q_j(n)$ (or simply q) is the number of odd terms among $n, T(n), \dots, T^{j-1}(n)$; the *coefficient* of $\Omega_j(n)$ is

$$C_j(n) = \frac{3^q}{2^j},$$

and $V_j(n) \in \{0, 1\}^j$ is the *parity vector* of length j induced by n , namely $V_j(n) = (n, T(n), \dots, T^{j-1}(n)) \bmod 2$. As in [1, Def. 2.1] we write parity vectors as binary words and abbreviate repetitions by superscripts, so that $1^2 0^3 1$ stands for $\langle 110001 \rangle$.

Date: July 11, 2026.

In collaboration with Claude Code. All results stated below are machine-verified in Lean 4; see Appendix A.

Definition 1.1 ([1, Def. 1.1]). For positive integers j and n with $n > 2$, the sequence $\Omega_j(n)$ is *paradoxical* if

- (i) $C_j(n) < 1$, and
- (ii) $T^j(n) \geq n$.

We write

$$\mathcal{P} = \{(j, n) : \Omega_j(n) \text{ paradoxical, } n > 2\}$$

for the *paradoxical set*.

Remark 1.2. In the formal development the two conditions (i)–(ii) constitute the paradoxicality predicate by themselves, and the side condition $n > 2$ is carried separately by every statement below; the conjunction agrees exactly with Definition 1.1. (The exclusion of sequences that merely repeat the trivial cycle, stipulated for convenience in [1, Def. 1.1], is automatic once $n > 2$.) The formal predicate is defined for all $j \geq 0$, but for $j = 0$ condition (i) fails since $C_0(n) = 1$, so nothing is added beyond the positive lengths considered in [1].

A paradoxical sequence with a unique local maximum is, in the cyclic case, classically called a 1-*cycle* or *circuit*; its parity vector is a single block $1^q 0^{j-q}$ [1, App. A]. We extend the word to the individual blocks of a general parity vector:¹ a *circuit* is a pair (a, e) of nonnegative integers, spelling the word $1^a 0^e$, and a finite list $S = ((a_1, e_1), \dots, (a_R, e_R))$ of circuits spells the concatenated word

$$w(S) = 1^{a_1} 0^{e_1} 1^{a_2} 0^{e_2} \dots 1^{a_R} 0^{e_R},$$

of length $\sum_{i=1}^R (a_i + e_i)$.

Definition 1.3. For $R \geq 0$, the R -*circuit slice* $\mathcal{P}_R \subseteq \mathcal{P}$ is the set of pairs $(j, n) \in \mathcal{P}$ such that

$$V_j(n) = 1^{a_1} 0^{e_1} \dots 1^{a_R} 0^{e_R}$$

for some nonnegative integers $a_1, e_1, \dots, a_R, e_R$ with $\sum_i (a_i + e_i) = j$.

Blocks in Definition 1.3 are allowed to be empty, so the slices are not disjoint (padding with the circuit $(0, 0)$ embeds $\mathcal{P}_R$ into $\mathcal{P}_{R+1}$); the results below concern only their union. We shall also use the following finiteness criterion from the formal development accompanying [1], which is proved there by bounding the first term of a paradoxical sequence of length j by $2^j 3^j$.

Proposition 1.4. *If there is a $J \in \mathbb{N}$ such that $j \leq J$ for every $(j, n) \in \mathcal{P}$, then $\mathcal{P}$ is finite.*

¹In [1, App. A] *circuit* refers to the whole (near-)cyclic sequence whose parity vector is one block $1^q 0^{j-q}$. We are not aware of standard terminology for the individual blocks and adopt *circuit* for them, as each block records one ascent–descent phase of the sequence.

2. THE CANONICAL CIRCUIT DECOMPOSITION

Every binary word is spelled by a canonical list of circuits, obtained by grouping each maximal block of 1's with the maximal block of 0's that follows it.

Definition 2.1. The *canonical circuit decomposition* $\sigma(w)$ of a finite binary word w is defined by recursion: $\sigma(\varepsilon) = \varepsilon$ for the empty word, and for $w \neq \varepsilon$,

$$\sigma(w) = ((a, e)) \frown \sigma(w''),$$

where 1^a is the maximal initial block of 1's of w , 0^e is the maximal block of 0's following it, and w'' is the remaining suffix. A word beginning with 0 thus yields a first circuit with empty ascent, $a_1 = 0$.

(The formal definition operates on words over $\mathbb{N}$, counting every letter different from 1 towards the block 0^e ; on binary words this is the description just given.) Since each recursion step consumes at least one letter, $\sigma(w)$ is nonempty whenever w is, and the recursion terminates.

Proposition 2.2. *For every finite binary word w , the canonical circuits of w spell w : if $\sigma(w) = ((a_1, e_1), \dots, (a_R, e_R))$, then*

$$w = 1^{a_1}0^{e_1} \dots 1^{a_R}0^{e_R}.$$

In particular $\sum_{i=1}^R (a_i + e_i)$ equals the length of w .

Proof sketch. Induction along the recursion of Definition 2.1. Because w is binary, the maximal initial block of 1's and the following maximal block of non-1's are the constant words 1^a and 0^e , and w is their concatenation with the suffix w'' . The inductive hypothesis applies to w'' (a suffix of a binary word being binary), and reassembling the three pieces recovers w . The length statement follows by comparing lengths on both sides. $\square$

3. COVERAGE: EVERY PARADOXICAL SEQUENCE LIES IN ITS CANONICAL SLICE

The parity vector $V_j(n)$ is a binary word of length j , so Proposition 2.2 applies to it.

Definition 3.1. The *canonical circuit count* $R_j(n)$ is the number of circuits of $\sigma(V_j(n))$.

For $j \geq 1$ the word $V_j(n)$ is nonempty and hence $R_j(n) \geq 1$.

Lemma 3.2 (Coverage). *If $\Omega_j(n)$ is paradoxical with $n > 2$, then $(j, n) \in \mathcal{P}_{R_j(n)}$.*

Proof sketch. The canonical decomposition $\sigma(V_j(n))$ consists of $R_j(n)$ circuits by definition, spells $V_j(n)$ by Proposition 2.2, and has total length j ; it therefore witnesses the membership required by Definition 1.3. $\square$

Corollary 3.3 (Exhaustion). *The circuit slices cover the paradoxical set exactly:*

$$\bigcup_{R \geq 0} \mathcal{P}_R = \mathcal{P}.$$

Proof sketch. Every member of every slice lies in $\mathcal{P}$ by Definition 1.3; conversely, Lemma 3.2 places each $(j, n) \in \mathcal{P}$ in the slice of its canonical circuit count. $\square$

4. THE UNIFORM-OVER- R CONNECTION

The formal development accompanying [1] proves, for each fixed $R \geq 1$, that the slice $\mathcal{P}_R$ is finite, conditionally on an excursion estimate of polynomial type imposed along the members of the slice — a windowed form of the bound $M_T(n) \leq n^\beta$ of [1, eq. (16)], which is derived there from Conjecture 6.2. Internally, that per- R theorem derives a bound $j \leq J(R)$ on the lengths occurring in $\mathcal{P}_R$. It is natural to ask what a single bound J valid for *all* R would give. The answer is: everything.

Theorem 4.1. *The following are equivalent.*

- (a) *There is a $J \in \mathbb{N}$ such that $j \leq J$ for every R and every $(j, n) \in \mathcal{P}_R$.*
- (b) *The paradoxical set $\mathcal{P}$ is finite.*

Proof sketch. (a) $\Rightarrow$ (b): by Lemma 3.2 every $(j, n) \in \mathcal{P}$ lies in some slice, so $j \leq J$ holds on all of $\mathcal{P}$, and Proposition 1.4 yields finiteness. (b) $\Rightarrow$ (a): if $\mathcal{P}$ is finite, the set of lengths occurring in $\mathcal{P}$ is finite, hence bounded by some J ; and every member of every slice belongs to $\mathcal{P}$, so $j \leq J$ on every slice. $\square$

Remark 4.2. By [1, Cor. 3.3], the finiteness of the paradoxical set implies the Collatz conjecture. Theorem 4.1 therefore makes precise the observation that a length bound uniform over all R “would be Collatz itself”: the uniform-over- R version of the per- R finiteness theorem is not merely sufficient for, but equivalent to, the finiteness of the paradoxical set.

Remark 4.3. We emphasize what is *not* claimed: assuming the excursion estimate of [1, Conj. 6.2] uniformly in R does not produce the uniform length bound of Theorem 4.1(a) through the per- R argument. That argument extracts a collision between circuits via a pigeonhole exponent of order $j/(2R)$; for a sequence with $R \approx j/2$ circuits this exponent degenerates to a constant and the collision disappears. This gap is exactly the Collatz-hardness of the uniform statement.

APPENDIX A. FORMAL STATEMENTS

The results of this note are machine-verified in Lean 4, in the file `paradoxical/CanonicalShape.lean` of the author’s formal development; all proofs are complete and free of extralogical axioms. The table below maps each environment to its formal counterpart. Statements recalled from companion files are marked accordingly.

Environment	Lean declaration	status
Def. 1.1	<code>RT.IsParadoxical</code>	definition (companion file)
Def. 1.3	<code>RCircuitSlice</code>	definition (companion file)
Prop. 1.4	<code>RT.finitely_many_paradoxical_of_</code> <code>paradoxical_length_bounded</code>	proved (companion file)
Def. 2.1	<code>Paradoxical.shapeOf</code>	definition
Prop. 2.2	<code>Paradoxical.wordOfShape_shapeOf</code>	proved
Def. 3.1	<code>Paradoxical.circuitCount</code>	definition
Lem. 3.2	<code>Paradoxical.rCircuitSlice_circuitCount</code>	proved
Cor. 3.3	<code>Paradoxical.iUnion_rCircuitSlice</code>	proved
Thm. 4.1	<code>Paradoxical.uniform_slice_length_</code> <code>bound_iff_finitely_many_paradoxical</code> (forward direction also separately: <code>Paradoxical.finitely_many_paradoxical_</code> <code>of_uniform_slice_length_bound</code>)	proved

REFERENCES

- [1] O. Rozier and C. Terracol, *Paradoxical behavior in Collatz sequences*, arXiv:2502.00948 (2025).